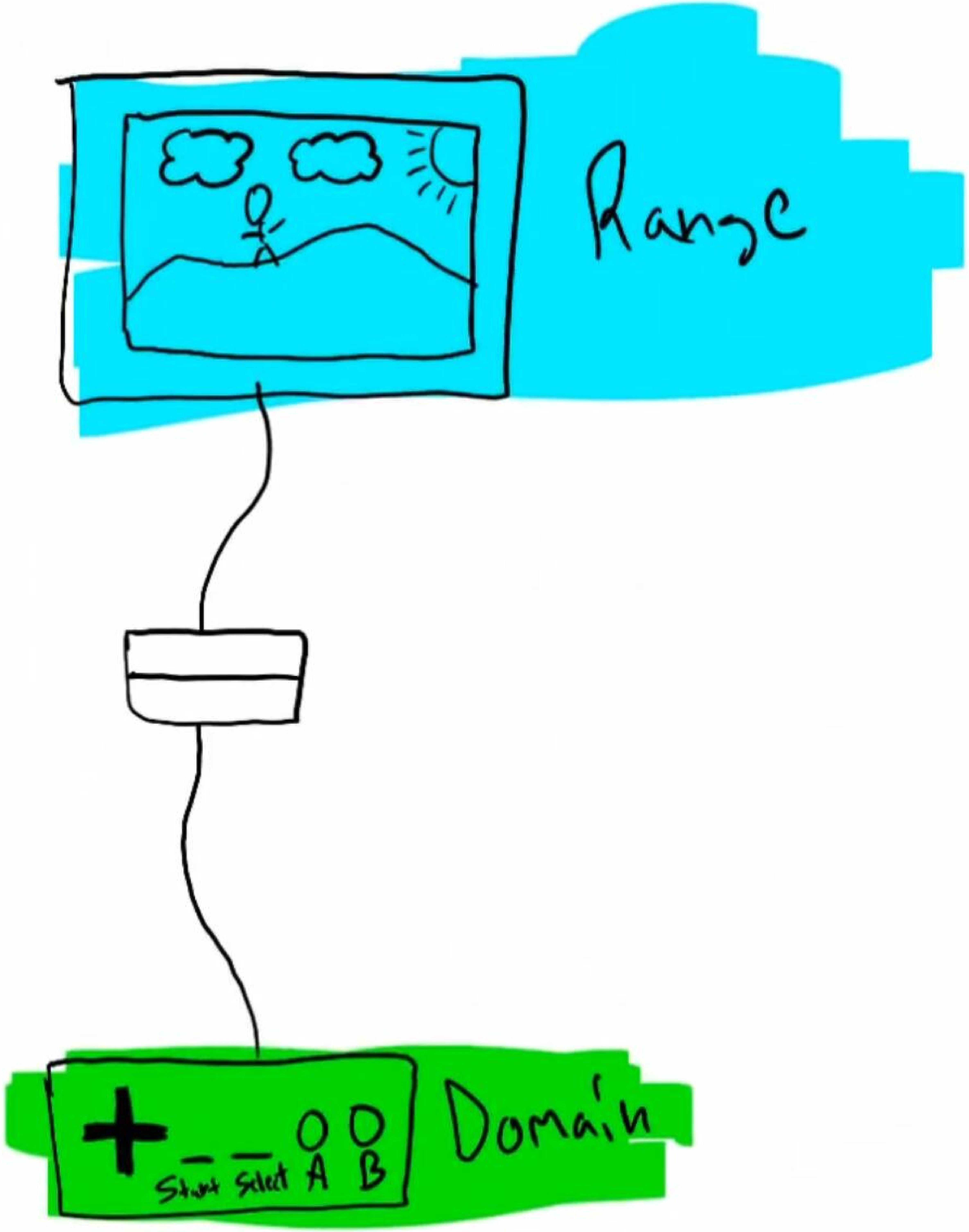
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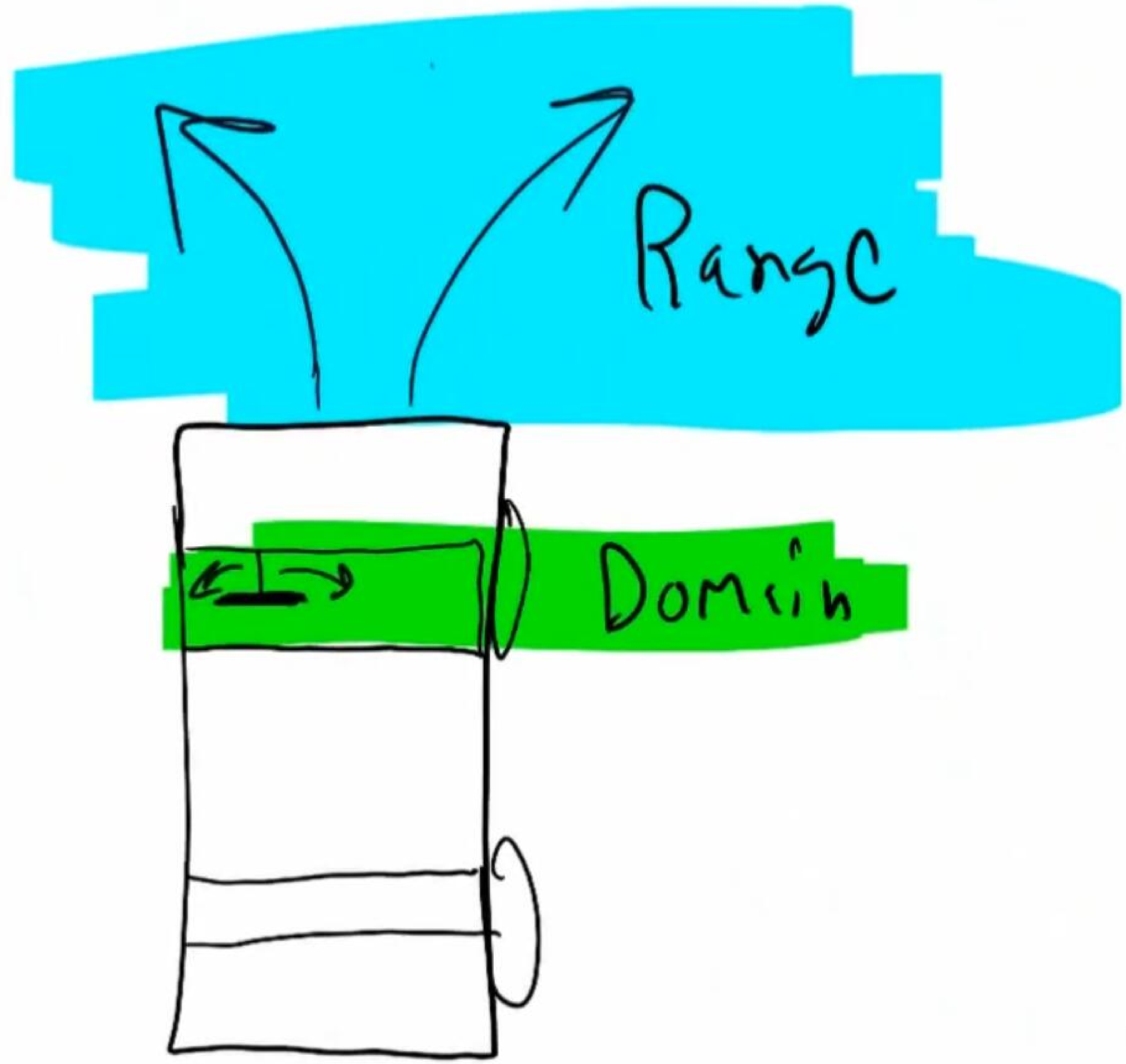
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Course

Functions and Function Notation (Part III)

The Domain of a Function

Domain: All of the inputs of a function (typically called x-values) that are able to produce an output.

* Neither imaginary nor complex numbers are allowed to be inputs or outputs of functions.

Find the domain of each function below and explain your reasoning.

① $f(x)=\frac{1}{x}$

Reasoning: A fraction with zero as its denominator is undefined.

Domain: All real numbers except zero.

② $g(x)=\frac{-7}{x-4}$

Reasoning: A fraction with zero as its denominator is undefined, so $x-4$ cannot be zero.

$$x-4\neq 0$$

$$x\neq 4$$

Domain: All real numbers except 4.

③ $h(x)=\frac{5}{2x+10}$

Reasoning: A fraction with zero as its denominator is undefined, so $2x+10$ cannot be zero.

$$2x+10\neq 0$$

$$2x\neq -10$$

$$x\neq -5$$

Domain: All real numbers except -5.

④ $f(x)=\frac{1}{x+10}$

Reasoning: A fraction with zero as its denominator is undefined, so $x+10$ cannot be zero.

$$x+10\neq 0$$

$$x\neq -10$$

Domain: All real numbers except -10.

⑤ $g(x) = \frac{-4}{3x}$

Reasoning: A fraction with zero as its denominator is undefined, so <u>3x</u> cannot be zero.

$$3x \neq 0$$

$$x \neq 0$$

Domain: All real numbers except zero.

⑥ $h(x) = \frac{2}{5-9x}$

Reasoning: A fraction with zero as its denominator is undefined, so <u>5−9x</u> cannot be zero.

$$5 - 9x \neq 0$$

$$-9x \neq -5$$

$$x \neq 5/9$$

Domain: All real numbers except 5/9.

Review on Inequalities

Example I: $3 > 2x - 1$

$$4 > 2x$$

$$\boxed{2 > x}$$

Example II: $x/3 + 5 \leq 2$

$$\frac{x}{-3} \leq -3$$

$$\boxed{x \geq 9}$$

Example III: $-5x + 6 \geq 16$

$$-5x \geq 10$$

$$\boxed{x \leq -2}$$

Example IV: $4 < 8 + x/4$

$$-4 < x/4$$

$$\boxed{-16 < x}$$

⑦ $f(x) = \sqrt{x}$

Reasoning: The square root of a negative number is an imaginary number, so x must be greater than or equal to zero.

Domain: All real numbers greater than or equal to zero.

⑧ $g(x) = \sqrt{5-x}$

Reasoning: The square root of a negative number is an imaginary number, so <u>5−x</u> must be greater than or equal to zero.

$$5 - x \geq 0$$

$$-x \geq -5$$

$$x \leq 5$$

Domain: All real numbers less than or equal to 5.

⑨ $h(x) = \sqrt{2x+1}$

Reasoning: The square root of a negative number is an imaginary number, so $\underline{2x+1}$ must be greater than or equal to zero.

$$2x+1 \geq 0$$

$$2x \geq -1$$

$$x \geq -\frac{1}{2}$$

Domain: All real numbers greater than or equal to $-\frac{1}{2}$.

⑩ $f(x) = \sqrt{-1+x}$

Reasoning: The square root of a negative number is an imaginary number, so $\underline{-1+x}$ must be greater than or equal to zero.

$$-1+x \geq 0$$

$$x \geq 1$$

Domain: All real numbers greater than or equal to 1.

⑪ $g(x) = \sqrt{-6x}$

Reasoning: The square root of a negative number is an imaginary number, so $\underline{-6x}$ must be greater than or equal to zero.

$$-6x \geq 0$$

$$x \leq 0$$

Domain: All real numbers less than or equal to zero.

⑫ $h(x) = \sqrt{5 - x/7}$

Reasoning: The square root of a negative number is an imaginary number, so $\underline{5 - x/7}$ must be greater than or equal to zero.

$$5 - \frac{x}{7} \geq 0$$

$$-\frac{x}{7} \geq -5$$

$$x \leq 35$$

Domain: All real numbers less than or equal to 35.

⑬ $f(x) = \log_3 x$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so x must be greater than zero.

Domain: All real numbers greater than zero.

(14) $g(x) = \log_5(x+7)$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so $\underline{\ x+7\ }$ must be greater than zero.

$$x+7>0$$

$$x>-7$$

Domain: All real numbers greater than -7.

(15) $h(x) = 3 - 4\log_8\left(4-\frac{x}{2}\right)$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so $\underline{\ 4-x/2\ }$ must be greater than zero.

$$4-\frac{x}{2}>0$$

$$-\frac{x}{2}>-4$$

$$x<8$$

Domain: All real numbers less than 8.

(16) $f(x) = \log_7(9x)$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so $\underline{\ 9x\ }$ must be greater than zero.

$$9x>0$$

$$x>0$$

Domain: All real numbers greater than zero.

(17) $g(x) = \log_4\left(x-\frac{1}{2}\right)$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so $\underline{\ x-1/2\ }$ must be greater than zero.

$$x-\frac{1}{2}>0$$

$$x>\frac{1}{2}$$

Domain: All real numbers greater than $1/2$.

(18) $h(x) = \log_3\left(-\frac{x}{4}+8\right)$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so $-x/4+8$ must be greater than zero.

$$-\frac{x}{4}+8>0$$

$$-\frac{x}{4}>-8$$

$$x<32$$

Domain: All real numbers less than 32.

(19) $f(x) = \begin{cases} x-3, & 4<x<8 \\ 2+x, & 9<x<13 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between 4 and 8 or between 9 and 13.

(20) $f(x) = \begin{cases} x^2-3, & -5<x<7 \\ 5x+1, & 10<x<23 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between -5 and 7 or between 10 and 23.

(21) $f(x) = \begin{cases} \frac{1}{x}, & -62<x<-40 \\ 4x, & 0<x<86 \end{cases}$

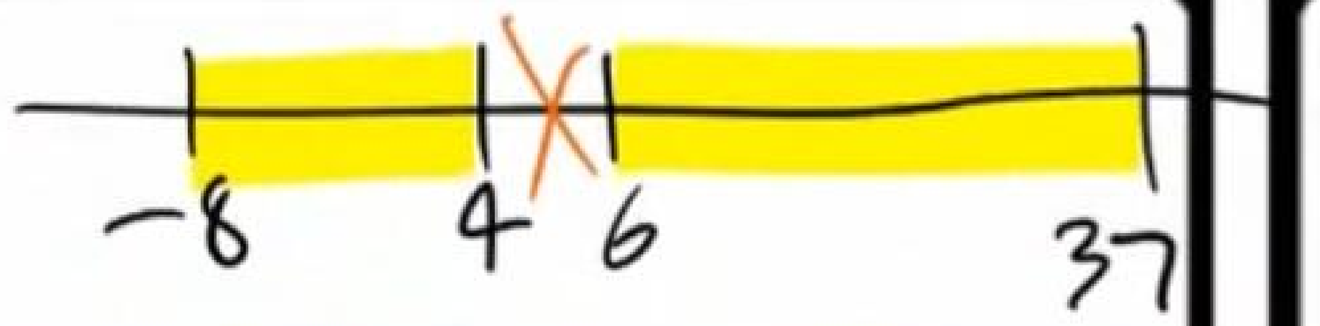
Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between -62 and -40 or between zero and 86.

(22) $f(x) = \begin{cases} 7^x, & 2<x<4 \\ x^2-2, & 7<x<12 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between 2 and 4 or between 7 and 12.



(23) $f(x) = \begin{cases} 2x-4, & -8 \le x \le 4 \\ 7-4x, & 6 \le x \le 37 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between -8 and 4, inclusive, or all real numbers between 6 and 37, inclusive.

(24) $f(x) = \begin{cases} 10^x, & 0 < x < 3 \\ x^3+1, & 3 < x \le 9 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between zero and 9, including 9.

(25) $f(x) = \begin{cases} 7, & 4 \le x \le 12 \\ 3, & 13 \le x \le 15 \end{cases}$

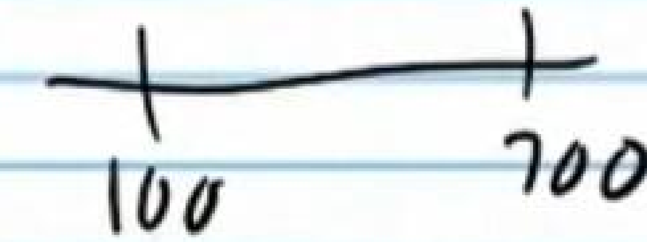
Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between 4 and 12, inclusive, or all real numbers between 13 and 15, inclusive.

(26) $f(x) = \begin{cases} 8x, & -4 \le x \le -1 \\ -1+x^3, & -1 < x < 6 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between -4 and 6, including -4.



(27) $f(x) = \begin{cases} 2x-1, & 100 < x < 300 \\ 4, & 300 \le x < 700 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between 100 and 700.

(28) $f(x) = \begin{cases} 5, & -10 < x \le -3 \\ -6, & -3 < x < 87 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between -10 and 87.

(29) $f(x) = \begin{cases} 3^{-x}, & -1 < x < 0 \\ 4^{x}, & 0 \leq x < 17 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between −1 and 17.

(30) $f(x) = \begin{cases} 9x^{2}, & 0 < x \leq 6 \\ -3x, & 6 < x < 14 \end{cases}$

Reasoning: This is a piecewise function, so the accepted inputs (i.e. the domain) are described with inequality notation alongside the function.

Domain: All real numbers between zero and 14.

Functions with No Domain Restrictions

(31) $f(x) = 3x + 6$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain: All real numbers.

(32) $g(x) = -4x^{2} + x - 7$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain: All real numbers.

(33) $f(x) = |x|$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain: All real numbers.

(34) $h(x) = x^{3}$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain: All real numbers.

(35) $g(x) = 2^{x+3}$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain: All real numbers.

Three Ways to Indicate Domain

I) Words

II) Set Notation

$\{x \mid x < 7\}$

The set of — all x — such that — $x<7$

III) Interval Notation

Uses parentheses, brackets, the union symbol, and the infinity symbol to describe intervals that represent possible values of the domain.

Example 1

Words	Set Notation	Interval Notation
All real numbers greater than 4.	$\{x \mid x>4\}$	$(4,\infty)$

Example 2

Words	Set Notation	Interval Notation
All real numbers greater than or equal to 4.	$\{x \mid x \geq 4\}$	$[4\ \infty)$

Example 3

Words	Set Notation	Interval Notation
All real numbers except 7.	$\{x \mid x \neq 7\}$	$(-\infty,7) \cup (7,\infty)$

Words	Set Notation	Interval Notation
All real numbers between −100 and −17.	$\{x \mid -100 < x < -17\}$	$(-100,-17)$

Example 5

Words	Set Notation	Interval Notation
All real numbers between −100 and −17, inclusive.	$\{x \mid -100 \leq x \leq -17\}$	$[-100,-17]$

Example 6

Words	Set Notation	Interval Notation
−8 only	$\{x \mid x=-8\}$	−8 is not an interval so interval notation is not useful in this example.

Example 7

Words	Set Notation	Interval Notation
All real numbers	All real numbers	$(-\infty,\infty)$

Interval Notation

Write the domain of each function using interval notation.

(36) $f(x) = \begin{cases} 2x-1, & 100 < x < 300 \\ 4, & 300 \le x < 700 \end{cases}$

Words: All real numbers between 100 and 700.

Interval Notation: $(100, 700)$

(37) $f(x) = \begin{cases} 8x, & -4 \le x \le -1 \\ -1+x^3, & -1 < x < 6 \end{cases}$

Words: All real numbers between −4 and 6, including −4.

Interval Notation: $[-4, 6)$

(38) $f(x) = \begin{cases} 10^x, & 0 < x \le 3 \\ x^3+1, & 3 < x \le 9 \end{cases}$

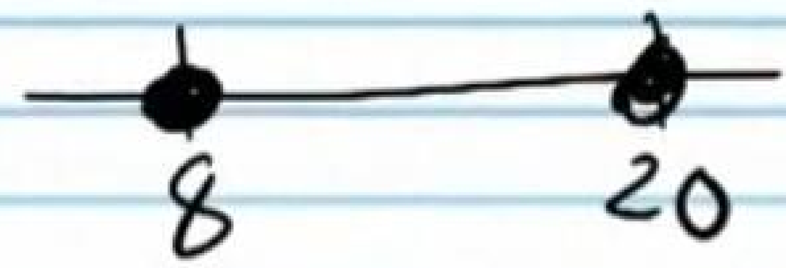
Words: All real numbers between zero and 9, including 9.

Interval Notation: $(0, 9]$

(39) $f(x) = \begin{cases} 7^x - 2x, & 8 \le x \le 10 \\ 4, & 10 < x \le 20 \end{cases}$

Words: All real numbers between 8 and 20, inclusive.

Interval Notation: $[8, 20]$



(40) $f(x) = \begin{cases} 5, & -10 < x \le -3 \\ -6, & -3 < x < 87 \end{cases}$

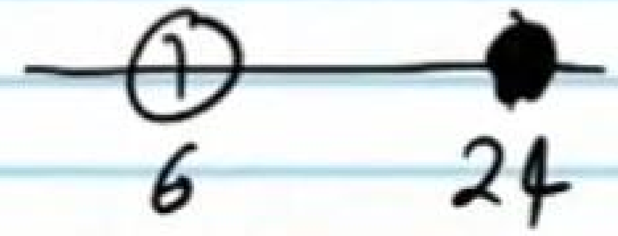
Words: All real numbers between −10 and 87.

Interval Notation: $(-10, 87)$

(41) $f(x) = \begin{cases} x^2-2x+4, & 6 < x < 15 \\ -8x^2-4x+9, & 15 \le x \le 24 \end{cases}$

Words: All real numbers between 6 and 24, including 24.

Interval Notation: $(6, 24]$



(42) $f(x) = \begin{cases} 2x^3+3^x, & 243 \le x \le 290 \\ 8-4x^2, & 290 < x < 384 \end{cases}$

Words: All real numbers between 243 and 384, including 243.

Intervals Notation: $[243, 384)$

(43) $f(x) = \begin{cases} 2x-3, & -7 \le x \le -5 \\ 8x-2, & -5 < x \le 36 \end{cases}$

Words: All real numbers between −7 and 36, inclusive.

Interval Notation: $[-7, 36]$

(44) $g(x) = \log_5(x+7)$

Words: All real numbers greater than -7.

Interval Notation: $(-7, \infty)$

(45) $g(x) = \sqrt{5-x}$

Words: All real numbers less than or equal to 5.

Interval Notation: $(-\infty, 5]$

(46) $h(x) = \log_3(-\frac{x}{4}+8)$

Words: All real numbers less than 32.

Interval Notation: $(-\infty, 32)$

(47) $f(x) = \sqrt{-1+x}$

Words: All real numbers greater than or equal to 1.

Interval Notation: $[1, \infty)$

(48) $h(x) = 2^{x+3}$

Words: All real numbers.

Interval Notation: $(-\infty, \infty)$

(49) $f(x) = \log_7(9x)$

Words: All real numbers greater than zero.

Interval Notation: $(0, \infty)$

(50) $h(x) = \sqrt{2x+1}$

Words: All real numbers greater than or equal to $-\frac{1}{2}$.

Interval Notation: $[-\frac{1}{2}, \infty)$

(51) $h(x) = 3 - 4\log_8(4-\frac{x}{2})$

Words: All real numbers less than 8.

Interval Notation: $(-\infty, 8)$

(52) $f(x) = \sqrt{x}$

Words: All real numbers greater than or equal to zero.

Interval Notation: $[0, \infty)$

(53) $f(x) = 3x+6$

Words: All real numbers.

Interval Notation: $(-\infty, \infty)$

(54) $f(x) = \begin{cases} x-3, & 4<x<8 \\ 2+x, & 9<x<13 \end{cases}$

Words: All real numbers between 4 and 8 or between 9 and 13.

Interval Notation: $(4,8) \cup (9,13)$

(55) $g(x) = \begin{cases} 5^x, & -8<x\le -3 \\ -x^2, & -2<x<-1 \end{cases}$

Words: All real numbers between -8 and -3, including -3, or all real numbers between -2 and -1.

Interval Notation: $(-8,-3] \cup (-2,-1)$

(56) $h(x) = \begin{cases} 3x^2+x-1, & -5\le x<3 \\ 2x^2+5x-10, & 8\le x<17 \end{cases}$

Words: All real numbers between -5 and 3, including -5, or all real numbers between 8 and 17, including 8.

Interval Notation: $[-5,3) \cup [8,17)$

(57) $f(x) = \begin{cases} 7, & 4\le x\le 12 \\ 3, & 13\le x\le 15 \end{cases}$

Words: All real numbers between 4 and 12, inclusive, or between 13 and 15, inclusive.

Interval Notation: $[4,12] \cup [13,15]$

(58) $f(x) = \begin{cases} x^2-3, & -5<x<7 \\ 5x+1, & 10<x<23 \end{cases}$

Words: All real numbers between -5 and 7 or between 10 and 23.

Interval Notation: $(-5,7) \cup (10,23)$

(59) $g(x) = \begin{cases} 7-x^3; & -12\le x<-10 \\ 2x, & 0<x<4 \end{cases}$

Words: All real numbers between -12 and -10, including -12, or all real numbers between zero and 4.

Interval Notation: $[-12,-10) \cup (0,4)$

(60) $h(x) = \begin{cases} |x|-2, & 5<x\le 7 \\ (-4)^x, & 9\le x<11 \end{cases}$

Words: All real numbers between 5 an 7, including 7, or all real numbers between 9 and 11, including 9.

Interval Notation: $(5,7] \cup [9,11)$

(61) $f(x) = \begin{cases} 2x-4, & -8\le x\le 4 \\ 7-4x; & 6\le x\le 37 \end{cases}$

Words: All real numbers between -8 and 4, inclusive, or between 6 and 37, inclusive.

Interval Notation: $[-8,4] \cup [6,37]$

62) $g(x) = \frac{-7}{x-4}$

Words: All real numbers except 4.

Interval Notation: $(-\infty,4) \cup (4,\infty)$

63) $h(x) = \frac{-4}{3x}$

Words: All real numbers except zero.

Interval Notation: $(-\infty,0) \cup (0,\infty)$

64) $h(x) = \frac{2}{5-9x}$

Words: All real numbers except 5/9.

Interval Notation: $(-\infty, 5/9) \cup (5/9,\infty)$

65) $f(x) = \frac{1}{x+10}$

Words: All real numbers except −10.

Interval Notation: $(-\infty,-10) \cup (-10,\infty)$

Set Notation

Write the domain of each function using set notation.

(66) $g(x) = \sqrt{5-x}$

Words: All real numbers less than or equal to 5.

Set Notation: $\{x \mid x \leq 5\}$

(67) $g(x) = \log_5(x+7)$

Words: All real numbers greater than −7.

Set Notation: $\{x \mid x > -7\}$

(68) $h(x) = \log_3\left(-\frac{x}{4}+8\right)$

Words: All real numbers less than 32.

Set Notation: $\{x \mid x < 32\}$

(69) $f(x) = \sqrt{-1+x}$

Words: All real numbers greater than or equal to 1.

Set Notation: $\{x \mid x \geq 1\}$

(70) $h(x) = 2^{x+3}$

Words: All real numbers.

Set Notation: All real numbers.

(71) $h(x) = \sqrt{2x+1}$

Words: All real numbers greater than or equal to $-\frac{1}{2}$.

Set Notation: $\{x \mid x \geq -\frac{1}{2}\}$

(72) $f(x) = \log_7(9x)$

Words: All real numbers greater than zero.

Set Notation: $\{x \mid x > 0\}$

(73) $h(x) = 3 - 4\log_8\left(4 - \frac{x}{2}\right)$

Words: All real numbers less than 8.

Set Notation: $\{x \mid x < 8\}$

(74) $f(x) = \sqrt{x}$

Words: All real numbers greater than or equal to zero.

Set Notation: $\{x \mid x \geq 0\}$

(75) $f(x) = 3x + 6$

Words: All real numbers.

Set Notation: All real numbers.

(76) $g(x) = \frac{-4}{3x}$

Words: All real numbers except zero.

Set Notation: $\{x \mid x \neq 0\}$

(77) $g(x) = \frac{-7}{x-4}$

Words: All real numbers except 4.

Set Notation: $\{x \mid x \neq 4\}$

(78) $h(x) = \frac{2}{5-9x}$

Words: All real numbers except 5/9.

Set Notation: $\{x \mid x \neq \frac{5}{9}\}$

(79) $f(x) = \frac{1}{x+10}$

Words: All real numbers except −10.

Set Notation: $\{x \mid x \neq -10\}$

(80) $f(x) = \begin{cases} 2x-1, & 100 < x < 300 \\ 4, & 300 \leq x < 700 \end{cases}$

Words: All real numbers between 100 and 700.

Set Notation: $\{x \mid 100 < x < 700\}$

(81) $g(x) = \begin{cases} 8x, & -4 \leq x \leq -1 \\ -1+x^3, & -1 < x < 6 \end{cases}$

Words: All real numbers between -4 and 6, including -4.

Set Notation: $\{x \mid -4 \leq x < 6\}$

(82) $f(x) = \begin{cases} 10^x, & 0 < x \leq 3 \\ x^3+1, & 3 < x \leq 9 \end{cases}$

Words: All real numbers between zero and 9, including 9.

Set Notation: $\{x \mid 0 < x \leq 9\}$

(83) $h(x) = \begin{cases} 7^x - 2x, & 8 \leq x \leq 10 \\ 4, & 10 < x \leq 20 \end{cases}$

Words: All real numbers between 8 and 20, inclusive.

Set Notation: $\{x \mid 8 \leq x \leq 20\}$

(84) $g(x) = \begin{cases} 5, & -10 < x \leq -3 \\ -6, & -3 < x < 87 \end{cases}$

Words: All real numbers between -10 and 87.

Set Notation: $\{x \mid -10 < x < 87\}$

(85) $f(x) = \begin{cases} x^2-2x+4, & 6 < x < 15 \\ -8x^2-4x+9, & 15 \leq x \leq 24 \end{cases}$

Words: All real numbers between 6 and 24, including 24.

Set Notation: $\{x \mid 6 < x \leq 24\}$

(86) $h(x) = \begin{cases} 2x^3+3^x, & 243 \leq x \leq 290 \\ 8-4x^2, & 290 < x < 384 \end{cases}$

Words: All real numbers between 243 and 384, including 243.

Set Notation: $\{x \mid 243 \leq x < 384\}$

(87) $f(x) = \begin{cases} 2x-3, & -7 \leq x \leq -5 \\ 8x-2, & -5 < x \leq 36 \end{cases}$

Words: All real numbers between -7 and 36, inclusive.

Set Notation: $\{x \mid -7 \leq x \leq 36\}$

(88) $f(x) = \begin{cases} x-3, & 4<x<8 \\ 2+x, & 9<x<13 \end{cases}$

Words: All real numbers between 4 and 8 or between 9 and 13.

Set Notation: $\{x \mid 4<x<8 \text{ or } 9<x<13\}$

(89) $g(x) = \begin{cases} 5^x, & -8<x\leq-3 \\ -x^2, & -2<x<-1 \end{cases}$

Words: All real numbers between -8 and -3, including -3, or all real numbers between -2 and -1.

Set Notation: $\{x \mid -8<x\leq-3 \text{ or } -2<x<-1\}$

(90) $h(x) = \begin{cases} 3x^2+x-1, & -5\leq x<3 \\ 2x^2+5x-10, & 8\leq x<17 \end{cases}$

Words: All real numbers between -5 and 3, including -5, or all real numbers between 8 and 17, including 8.

Set Notation: $\{x \mid -5\leq x<3 \text{ or } 8\leq x<17\}$

(91) $f(x) = \begin{cases} 7, & 4\leq x\leq 12 \\ 3, & 13\leq x\leq 15 \end{cases}$

Words: All real numbers between 4 and 12, inclusive or between 13 and 15, inclusive.

Set Notation: $\{x \mid 4\leq x\leq 12 \text{ or } 13\leq x\leq 15\}$